

# ME2801: Homework 1

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## 1 Nise 2.2

a.  $f(t) = e^{-at} \sin(\omega t) u(t)$

From Table 2.2, use Item no. 4, “Frequency shift theorem”. From Table 2.1, use Item no. 6.

$$F(s) = \frac{\omega}{(s+a)^2 + \omega^2}$$

b.  $f(t) = e^{-at} \cos(\omega t) u(t)$

$$F(s) = \frac{s+a}{(s+a)^2 + \omega^2}$$

c.  $f(t) = t^3 u(t)$

Could do this at least two ways. 1) Use Item no. 4 in Table 2.1 directly or 2) use the “Integration theorem” from Table 2.2

$$F(s) = \frac{6}{s^4}$$

## 2 Given $f(t)$ , find $F(s)$

Note that the unit step ( $u(t)$ ) is implied for each of these.

a.  $f(t) = 1 + 2t$

Applying the Laplace transform pairs in Table 2.1 to each term directly...

$$\begin{aligned} F(s) &= \frac{1}{s} + 2 \frac{1}{s^2} \\ &= \frac{s+2}{s^2} \end{aligned}$$

**b.**  $f(t) = 3 + 7t + t^2 + \delta(t)$

Again, applying the Laplace transform pairs, term-by-term...

$$F(s) = \frac{3}{s} + \frac{7}{s^2} + \frac{2}{s^3} + 1$$

**c.**  $f(t) = e^{-t} + 2e^{-2t} + te^{-3t}$

For the first two terms, we can use the Laplace pairs directly. For the third term we can apply the Frequency shift theorem (Item no. 4. in Table 2.2) and the Laplace transform pair no. 3 from Table 2.1.

$$\mathcal{L}[te^{-3t}] = \frac{1}{(s+3)^2}$$

Now...

$$F(s) = \frac{1}{s+1} + 2\frac{1}{s+2} + \frac{1}{(s+3)^2}$$

**d.**  $f(t) = 3\cos(6t)$

Using the Laplace pair for  $\cos(\omega t)$  and the Linearity theorem...

$$F(s) = 3\frac{s}{s^2 + 6^2}$$

**e.**  $f(t) = 2\cos(2t) + e^{-t}\sin(2t)$

For the second term, use the Frequency shift theorem as you did in Problem 1)

$$F(s) = 2\frac{s}{s^2 + 2^2} + \frac{2}{(s+1)^2 + 2^2}$$

## 3 Laplace transform of decaying sinusoids

### 3.1 Laplace Transform

**a.**  $f(t) = 10.0 (e^{-0.5t}\sin(2\pi t))$

Using results from Problem 1...

$$\begin{aligned} F(s) &= 10.0 \frac{2\pi}{(s+0.5)^2 + (2\pi)^2} \\ &= 10.0 \frac{2\pi}{s^2 + s + ((2\pi)^2 + 0.25)} \end{aligned}$$

**b.**  $g(t) = 10.0 (e^{-0.75t} \sin(2\pi t))$

Similarly...

$$\begin{aligned} G(s) &= 10.0 \frac{2\pi}{(s + 0.75)^2 + (2\pi)^2} \\ &= 10.0 \frac{2\pi}{s^2 + s + ((2\pi)^2 + (0.75)^2)} \end{aligned}$$

**c.**  $h(t) = 10.0 (e^{-0.5t} \sin(4\pi t))$

Similarly...

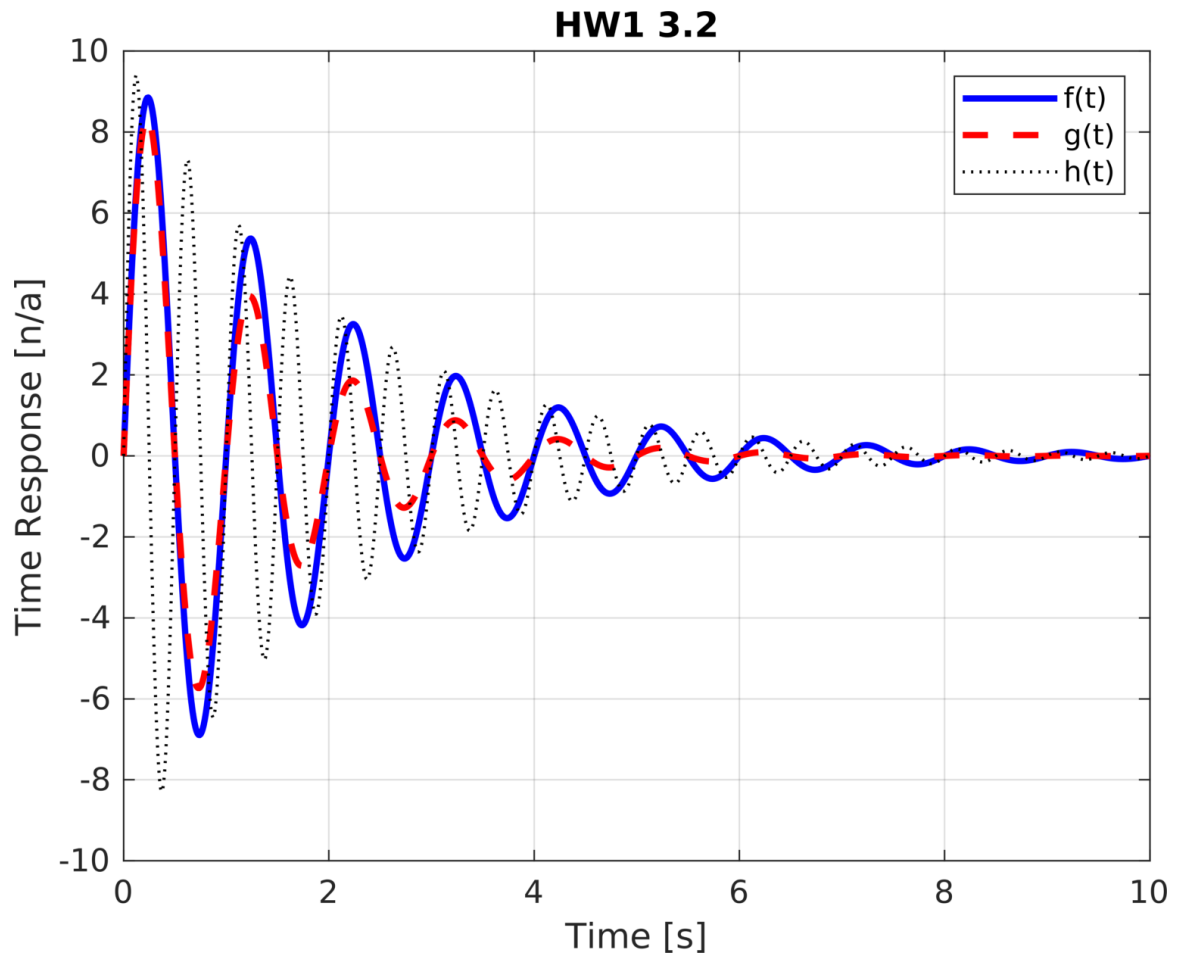
$$\begin{aligned} H(s) &= 10.0 \frac{4\pi}{(s + 0.5)^2 + (4\pi)^2} \\ &= 10.0 \frac{4\pi}{s^2 + s + ((4\pi)^2 + (0.5)^2)} \end{aligned}$$

### 3.2 Time-Domain Plot

% HW1, 3.2)

```
t = linspace(0,10,1000);
f = 10*exp(-0.5*t).*sin(2*pi*t);
g = 10*exp(-0.75*t).*sin(2*pi*t);
h = 10*exp(-0.5*t).*sin(4*pi*t);
```

```
figure(1);
clf();
plot(t,f,'b','linewidth',2);
hold('on')
plot(t,g,'r--','linewidth',2);
plot(t,h,'k:', 'linewidth',1);
title('HW1 3.2');
ylabel('Time Response [n/a]');
xlabel('Time [s]');
legend('f(t)', 'g(t)', 'h(t)');
grid('on')
```



**4 Given  $F(s)$ , find  $f(t)$ , using MATLAB for partial fraction expansion**

**a.**  $F(s) = \frac{2}{s(s+2)}$

Using MATLAB for the PFE

```
% HW1, 4) a)
num = 2;
den = [1 2 0];
[R,P,K] = residue(num,den)
```

```

R =
    -1
     1
P =
    -2
     0
K =
    []

```

Which says we can rewrite...

$$F(s) = \frac{-1}{s+2} + \frac{1}{s}$$

You might verify that this is the correct expression for  $F(s)$  by finding the common denominator...

$$F(s) = \frac{-2}{s(s+2)} + \frac{(s+2)}{s(s+2)} = \frac{2}{s(s+2)}$$

Use the Laplace transform pairs (Table 2.1) to find the inverse Laplace transform of each term...

$$\boxed{f(t) = (-1(e^{-2t}) + 1) u(t)}$$

where the explicitly including the  $u(t)$  is optional.

**a.**  $F(s) = \frac{10}{s(s+1)(s+10)}$

Using MATLAB for the PFE...

```

% HW1, 4) b)
num = 10;
den = conv([1 0],conv([1 1],[1 10]));
[R,P,K] = residue(num,den)

```

```

R =
    0.1111
   -1.1111
    1.0000
P =
   -10

```

$$K = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Interpreting the MATLAB output as...

$$F(s) = \frac{0.11}{s+10} + \frac{-1.11}{s+1} + \frac{1.0}{s}$$

Now take the inverse Laplace transform of each term...

$$\boxed{f(t) = [0.11e^{-10t} - 1.11e^{-t} + 1.0] u(t)}$$